

JEE MAIN REVISION MASTER: BIOMOLECULES

2026 3 QUESTIONS 5% WEIGHTAGE

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 27 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Kcet Biomolecules - Core Application	27 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Kcet Biomolecules - Core Application (Very High Yield — 27 PYQs)

Core Principle: Key problem pattern in Kcet Biomolecules.

■ ■ Caution / JEE Pitfall: Watch for algebraic and sign errors.

■ Speed Trick: Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "Incorrect statement about α -amino acids of proteins among the following is..." [Ans: N/A]
- [2026 Shift] Q#2 (MCQ): "Consider the following statements: Statement-I: All monosaccharides are reducing sugars. Statement-II: Sucrose can reduce ammoniacal silver ..." [Ans: N/A]
- [2026 Shift] Q#3 (MCQ): "Match List-I with List-II List-I (Vitamins) List-II (Deficiency Diseases) (a) B₁ (i) Convulsions (b) B₂ (ii) RBC def..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

Incorrect statement about α -amino acids of proteins among the following is

- (A) Methionine is an essential amino acid
- (B) Glycine doesn't exhibit enantiomerism
- (C) Glycylalanylglycine has three amide linkages
- (D) Zwitterion of valine exhibits amphoteric behaviour

Correct Answer: None

Solution Analysis: The incorrect statement is: Option C Glycylalanylglycine is a tripeptide made from three amino acid units: Glycine-Alanine-Glycine. In a peptide, two amino acids join by one peptide bond, also called an amide linkage. So, for three amino acid units, the number of peptide bonds is: $3-1=2$. Hence, glycylalanylglycine has only two amide linkages, not three. The other statements are correct: Methionine is an essential amino acid. Glycine does not show enantiomerism because its α -carbon has two hydrogen atoms, so it is not chiral. Valine exists as a zwitterion.